Rapid Flood Extent Mapping from Sentinel-1 SAR: A Controlled Study of Multi-Temporal U-Nets on Kuro Siwo

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***Abstract*—Synthetic aperture radar (SAR) maps floodwater through cloud and at night, and a pre-event image makes changes easy to isolate, but it is unclear how many pre-event images are needed and whether elaborate multi-temporal architectures pay off on unseen flood events. We study rapid flood mapping on Kuro Siwo with Sentinel-1 VV/VH images from two pre-event dates and one post-event date, using strictly event-disjoint training, validation and test sets. We propose a multi-temporal U-Net with a gated, gap-aware temporal fusion (days between acquisitions as input) and temporal dropout, and compare it with change-detection rules, post-event-only and early-fusion U-Nets, a difference-fusion variant, and a temporal-attention variant, with three seeds each, ablations on terrain (DEM) channels, speckle filtering and training recipes, and two recent pretrained baselines (U-Net with ResNet-50 encoder, SegFormer-B2), a cross-dataset evaluation against Sen1Floods11, input-attribution analyses, and one pre-registered evaluation on 10,839 patches of ten unseen events. A single pre-event image raises validation IoU from 0.532 to 0.684, whereas a second one, attention, gating, terrain channels and speckle filters do not help. On the test set the proposed model matches the plain U-Net (ensemble IoU 0.597 vs. 0.599; difference -0.003, 95% bootstrap interval $[-0.008, 0.003]$) although it was 0.014 IoU behind on validation. The gate learns to ignore the older image, because it is older than the first in every patch. About 40% of false-positive pixels come from flood-free patches, and permanent water causes 16–32%. The pretrained baselines gain about 0.01 IoU on validation but are indistinguishable from the plain U-Net on the unseen test events (0.583–0.587 single, 0.589–0.606 ensembles). Post-event-only models transfer poorly between datasets (IoU 0.23 from Kuro Siwo to Sen1Floods11, 0.37 in the other direction on the first test shard against 0.60 in-domain), and Shapley analysis shows that the post-event VV channel carries 68–77% of the explained IoU. We report a negative but carefully controlled result that points to data shift and flood-free patches, not model capacity, as the bottleneck.**

***Index Terms*—flood mapping, Sentinel-1, SAR, U-Net, multi-temporal, change detection, Kuro Siwo, Sen1Floods11, cross-dataset, explainability**

I. Introduction

Floods are among the most damaging natural hazards, and emergency response needs flood extent maps within hours. Optical imagery is often unusable during the storms that cause floods; SAR is largely unaffected by cloud and daylight, so Sentinel-1 is widely used for rapid mapping [1]. Open water is dark in SAR because of specular reflection, but permanent water, smooth surfaces and radar shadow look similar, which is why change with respect to a pre-event image is informative. Deep networks have replaced thresholding on several benchmarks [2]–[4], and recent work uses Siamese networks, attention and transformers on bitemporal SAR [5]–[7].

Research gap. Most studies evaluate on random or location-mixed splits, or on a few events, and report one model per architecture. To our knowledge, a controlled comparison under event-disjoint splits that isolates (i) whether pre-event imagery matters, (ii) whether *two* pre-event images are better than one, (iii) whether multi-temporal architectures and attention beat early fusion, and (iv) how the answers change when the test events have much longer pre-to-post gaps and many flood-free patches than the training events, is missing from the literature we reviewed. The Kuro Siwo benchmark paper reports that most segmentation models do best with all available pre-event images and that DEM inputs bring negligible gains [4]; we test these statements with repeated runs.

Objectives. (1) Build a reproducible, leakage-free pipeline for flood segmentation from two pre-event and one post-event Sentinel-1 observations. (2) Propose a multi-temporal U-Net whose fusion learns *how much to trust the older pre-event image*, given the acquisition gaps. (3) Compare it with strong simple baselines and ablations, with multiple seeds and confidence intervals. (4) Analyze failures by event, acquisition gap, flood-free patches and permanent water. (5) Test whether conclusions hold against recent pretrained baselines and on a second dataset, and explain what the models use.

Contributions. (a) A gated, gap-aware temporal fusion with temporal dropout and its comparison with difference fusion and temporal attention. (b) A controlled study on 25 Kuro Siwo events with a pre-registered final evaluation on ten unseen events. (c) Evidence that one pre-event image captures nearly all the gain, and that the second image, DEM channels and speckle filters add nothing measurable. (d) A failure analysis that attributes most remaining error to flood-free patches and small floods rather than to model capacity. (e) A cross-dataset evaluation on Sen1Floods11 with permanent-water-aware scoring, a comparison with two recent pretrained baselines, and an attribution study (Grad-CAM, Shapley values over input channels, gate maps).

II. Related Work

Flood datasets. Sen1Floods11 provides hand-labelled Sentinel-1/2 flood chips [2]; MMFlood provides a multimodal

flood delineation dataset [3]; SpaceNet 8 targets flooded roads and buildings [8]; WorldFloods supports onboard mapping [9]; UrbanSARFloods offers Sentinel-1 SLC chips for urban and open areas [10]. Kuro Siwo gives pre- and post-event Sentinel-1 patches with three-class labels for 43 events, and its BlackBench baselines include U-Net variants, a ConvLSTM, ChangeFormer-style and ViT-based models [4].

Deep learning on SAR floods. Yadav *et al.* use an attentive dual-stream Siamese U-Net on multi-temporal Sentinel-1 and report a 6% IoU gain over the best uni-temporal method on Sen1Floods11 [5]. DAM-Net and PDCA-Former apply differential-attention and cross-attention transformers to bitemporal SAR flood detection [6], [7]. Misra *et al.* map floods globally from ten years of Sentinel-1 data [11]. A recent review notes that vegetated and urban areas remain difficult [1].

Change detection. Remote-sensing change detection uses Siamese encoders with transformers [12], [13] and state-space models [14]. These assume two dates; we ask what a third (older) date can add.

Terrain and speckle. EvaNet injects elevation through an elevation-based loss and gated convolution [15]. Median or BM3D filtering raised water-segmentation F1 by only about 0.01 in a study of Sentinel-1 images [16].

Foundation models. Geospatial pre-trained models such as Prithvi, SatMAE, SSL4EO-S12 models, DOFA and TerraMind [17]–[22] are being adapted to flood mapping; on Sen1Floods11 a small Presto model reaches F1 0.718 from SAR alone versus 0.896 with SAR plus multispectral input [23], and Prithvi-CAFE reports IoU 83.41 against 82.50 for Prithvi on the test split [24]. SAR-only, event-disjoint evaluation with temporal gaps is less explored.

Explainability. Grad-CAM, Grad-CAM++ and SHAP are the usual tools [25]–[27]; Sanderson *et al.* use Grad-CAM to compare Sentinel-1 and Sentinel-2 fusion for flood mapping [28], and IMAFD proposes an interpretable multi-stage flood detector [29].

III. Dataset and Pre-processing

A. Kuro Siwo

Kuro Siwo [4] provides 224×224 patches (10 m) with two pre-event and one post-event Sentinel-1 GRD observation (VV, VH), a three-class mask (background, permanent water, flood), a valid-pixel mask and metadata with event IDs and acquisition dates. We use the 25 events of the training split and three shards of the test split, processing one large shard at a time.

B. Sen1Floods11

Sen1Floods11 [2] provides 446 hand-labelled 512×512 chips from 11 regions with one Sentinel-1 image (VV, VH, in dB) per chip, a water label (water includes permanent water) and the JRC permanent-water layer [30]. We use the official split: 252 training, 89 validation, 90 test chips from 10 regions, and 15 chips of Bolivia that form a held-out region. Mean water fractions are 9.3% (train), 11.0% (validation), 12.3% (test) and 17.0% (Bolivia); 35% of all chips have less than 1% water. The chips have no pre-event image, so only post-event models apply. Post-event dB percentiles are close to those of Kuro Siwo (VV median -9.8 vs. -8.9 dB, VH -16.3 vs. -15.7 dB), so the shift between the datasets is mainly in labels and scenes, not in calibration (Fig. 1). We apply the same dB conversion, clipping and the Kuro Siwo channel statistics to both datasets.

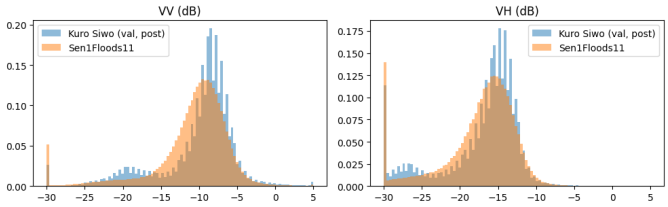


Fig. 1. Post-event backscatter histograms (dB) of Kuro Siwo (validation) and Sen1Floods11.

TABLE I
Splits. Gap = median days from pre-event 1 to the event.

Split	Patches	Events	Flood-free	Gap (d)
Train	7,925*	22	14%	24
Validation	3,684	3	27%	48
Test shard 0 (explor.)	6,000	9	69%	168
Fresh test (shards 1, 2)	10,839	10	–	180–240†

*plus 1,160 extra flood-free patches; †80% of patches.

C. Splits (event-disjoint)

Splits are by flood event, never by patch (Table I). Validation uses three training-split events; a cap of 1,500 random patches per event limits the dominance of the largest events. The fresh test uses shard 1 and the eligible events of shard 2; events 470 and 492, which also occur in training/validation, are excluded (1,161 patches). Test shard 0 was used for exploratory analysis only. The test differs strongly from training: 80% of its patches have a gap of 180–240 days between the closer pre-event image and the event (training median: 24 days) and most patches are flood-free. Patch coordinates are in a local projected system without a zone, so we claim generalization to unseen *events*, not to unseen regions.

D. Pre-processing

Linear backscatter is converted to dB, clipped to $[-30, +5]$ dB, stored as 8-bit and standardized per channel with training-set statistics. The target is the flood class (label 2); permanent water and background are the negative class, and invalid pixels are excluded from loss and metrics. Speckle is not filtered in the main experiments (Lee and median filters are evaluated as ablations, Sec. V-B). Augmentation uses random flips and 90° rotations; the data are provided as 224×224 tiles, and we add a tiling-and-blending routine for full scenes (overlapping tiles, Hann-weighted blending), verified to reproduce patch-wise predictions (max. probability difference $< 10^{-3}$).

IV. Methods

A. Baselines

Change detection: a threshold on the VV post-minus-pre difference (-3 dB), and a post-event VV threshold (-14 dB), both fitted on training data. **U-Net** [31]: four-level encoder–decoder with base width 32 (7.76 M parameters), early fusion of 2 (post only), 4 (pre1 + post) or 6 (pre1, pre2, post) channels. **Multi-temporal (shared encoder) variants:** `mt_diff` (one encoder over three dates; fusion at each scale by concatenating post, post minus mean pre, mean pre, then a 1×1 convolution) and `mt_attn` (softmax-weighted mix of the two pre-event features by similarity to the post features, plus additive attention gates [32]). **Recent pretrained baselines** (4 input channels, pre1 + post): a U-Net decoder on an ImageNet-pretrained ResNet-50 encoder [33] (28.0 M parameters), the best model family of the Kuro Siwo benchmark [4], and SegFormer-B2 [34], a hierarchical transformer with an all-MLP decoder (27.4 M parameters; the first patch embedding is extended to four channels). Both use the training recipe of the plain U-Net with learning rates fixed in advance (2×10^{-4} and 6×10^{-5}) and three seeds.

B. Proposed: gated, gap-aware multi-temporal U-Net

The model (Fig. 2) encodes pre1, pre2 and post with one shared encoder. At every scale, the reference feature is $\mathrm{ref} = f_1 + g\,(f_2 - f_1)$, where $g \in (0, 1)$ is a per-pixel gate $g = \sigma\big(b + s_2 - s_1 + \mathbf{w}^\top \boldsymbol{\ell}\big)$. Here s_k is the scaled dot product between a projection of the post feature and of pre feature k, $\boldsymbol{\ell}$ holds the standardized log-days from each pre-event image to the post image, and $b = -3$ so that the model starts as a plain pre1-versus-post change model and uses pre2 only if it helps. The fused feature is a 1×1 convolution of $[f_{\mathrm{post}},\ f_{\mathrm{post}} - \mathrm{ref},\ \mathrm{ref}]$. *Temporal dropout* removes one pre-event image with probability 0.3 during training, so the model sees long-gap situations. The decoder is a standard U-Net decoder (8.05 M parameters).

C. Training and inference

All plain U-Nets use AdamW (lr 3×10^{-4}, weight decay 10^{-4}), cosine schedule, 12 epochs, batch size 16, early stopping (patience 5), loss = masked BCE + Dice. The multi-temporal models use a stabilised recipe (lr 10^{-4}, weight decay 10^{-2}, EMA weights, bottleneck dropout 0.1, 14 epochs, patience 6). Hyperparameters were set a priori and not tuned on the results. Three seeds are trained per model; ensembles average sigmoid outputs of the three seeds. A fixed post-processing rule sets patches with fewer than 500 predicted flood pixels to "no flood". Variants of the loss (Tversky [35] with false alarms weighted more), hard-negative sampling of flood-free patches, shift augmentation and EMA were tested as an improvement round.

D. Evaluation protocol

Metrics are flood-class IoU, F1, precision and recall, pooled over valid pixels, since per-patch IoU is undefined for flood-free patches. We report mean $\pm$ standard deviation over three seeds and ensemble results, per-event and per-gap tables, a patch-level bootstrap for model differences (optimistic, as patches in an event are correlated), and an "IoU if flood-free patches had no false positives" column. Decision rules were fixed before seeing results: a variant replaces the baseline only if its mean single-model IoU is at least 0.015 higher (about two seed standard deviations) and its IoU with flood-free patches re-weighted to 69% is not lower; an ensemble is adopted if it adds at least 0.005. The final test was frozen on validation evidence and evaluated once; the pretrained baselines were added afterwards and are evaluated on that test only post hoc, as exploratory evidence.

TABLE II
Validation (3 events, 3 seeds, mean $\pm$ sd). Ens. = 3-seed ensemble (IoU).

Model	Params	IoU	F1	Ens.
Change detection (VV, -3 dB)	–	0.576	0.731	–
U-Net, post only (2 ch)	7.76 M	0.532 ± 0.003	0.695	–
U-Net, pre1 + post (4 ch)	7.76 M	$\mathbf{0.684 \pm 0.008}$	0.812	**0.692**
U-Net, pre1 + pre2 + post (6 ch)	7.76 M	0.668 ± 0.005	0.801	–
`mt_diff`	7.75 M	0.666 ± 0.010	0.799	–
`mt_attn`	8.13 M	0.665 ± 0.001	0.798	–
`mt_g2` (proposed)	8.05 M	0.669 ± 0.004	0.802	0.678

Cross-dataset protocol. Only a post-event image is shared by the two datasets, so we train a post-event-only U-Net (2 input channels) on each dataset (3 seeds) and test it in both directions. The Sen1Floods11 model is trained on random 224×224 crops of the training chips (water label, early stopping on the validation chips); chips are predicted at 512×512 with overlapping tiles. Because Sen1Floods11 "water" includes permanent water while Kuro Siwo "flood" excludes it, the like-for-like score (*flood_noperm*) counts water pixels against the temporary-flood target and ignores permanent-water pixels (JRC layer, Kuro Siwo class 1). We additionally report a *water* score, where the Kuro Siwo model's prediction is united with the JRC layer.

V. Results

A. Validation

Table II and Fig. 3 summarise three-seed validation results. Pre-event imagery is the key ingredient: a U-Net with one pre-event image reaches IoU 0.684 ± 0.008 against 0.532 ± 0.003 for post-event only ($+0.15$), and the change-detection rule reaches 0.576. A second pre-event image does not help (0.668), and neither do the shared-encoder multi-temporal models (0.665–0.669) nor the proposed gated model (0.669 ± 0.004), despite 2.7 times the training time. Ensembling three seeds adds $+0.008$ (0.692 for the plain U-Net, 0.678 for the proposed model).

B. Ablations

Improvement round. Explicit change channels (0.683), Tversky loss with hard negatives, shift augmentation and EMA (0.674), and both (0.668) did not beat the plain U-Net (0.684);

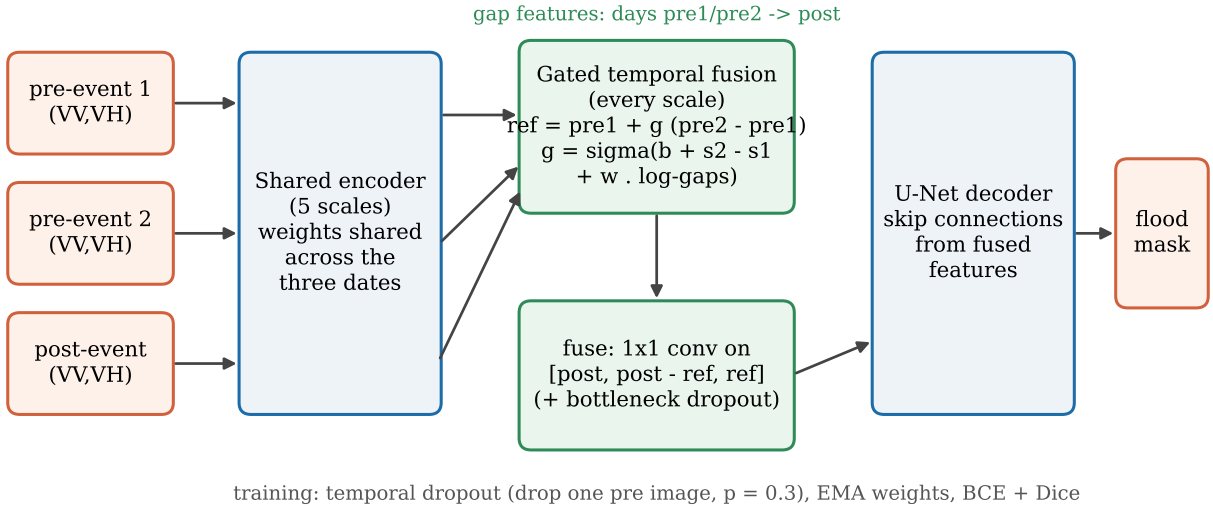


Fig. 2. Gated, gap-aware multi-temporal U-Net. A single encoder processes the three dates; at every scale a learned gate mixes the two pre-event features using similarity to the post-event feature and the acquisition gaps; temporal dropout randomly removes one pre-event image during training.

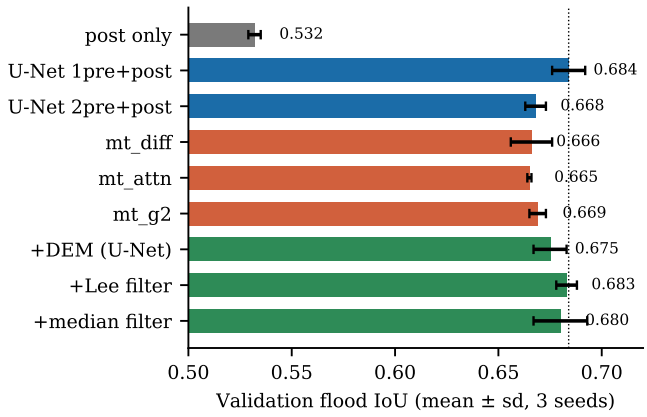


Fig. 3. Validation IoU of all models and ablations (3 seeds; error bars = sd). The dotted line marks the plain U-Net.

none reached the $+0.015$ rule. Test-time augmentation added $+0.001$ to an ensemble and was dropped.

Terrain. Adding patch-relative relief and slope (from the dataset's DEM) lowered validation IoU to 0.675 ± 0.008 for the plain U-Net and to 0.658 (one seed) for the multi-temporal model. This agrees with the benchmark paper's finding of negligible DEM gains [4].

Speckle filtering. A Lee-type filter (5×5, linear intensity) gave 0.683 ± 0.005 and a 3×3 median filter 0.680 ± 0.013, neither better than no filter (0.684), so we conclude the network learns its own smoothing; this is consistent with the small effects reported for median and BM3D filtering in [16].

Gate behaviour. The mean gate stays between 0.08 (fine scales) and 0.26 (coarsest scale), i.e. the model uses the older image little (Fig. 5, left). It uses it *less* when the closer pre-event image is old, the opposite of what we hoped. The reason is structural: the second pre-event image is older than the first in 100.0% of train and validation patches, so it is never the better reference.

C. Fresh test on unseen events

Table III gives the single pre-registered evaluation. Both models fall from about 0.69 on validation to about 0.60. The multi-temporal model, 0.014 behind on validation, is now on par with the plain U-Net (difference -0.003, bootstrap 95% interval $[-0.008, +0.003]$). The 500-pixel rule fixed in advance helps on the test as well ($+0.006$ for ensembles). Per-event IoU ranges from 0.24 to 0.91 (Fig. 4), a spread far larger than any difference between the models; the two worst events (279 and 427) hold 71% of the test patches and have mean flood fractions of 0.7% and 0.5%.

D. Recent pretrained baselines

Table IV compares the pretrained baselines with the plain and the proposed U-Net. On validation both pretrained models are about 0.01 IoU better than the plain U-Net (0.693 ± 0.007 and 0.696 ± 0.003 against 0.684 ± 0.008), below our pre-registered $+0.015$ rule, and the best ensemble (U-Net/ResNet-50) reaches 0.703. On the fresh test, evaluated post hoc, the four models are indistinguishable: single-model IoU is 0.583–0.587 with seed standard deviations up to 0.021, and ensembles range from 0.589 to 0.606. Per-event IoU again varies from 0.22 to 0.91 and the same events (279, 427) are worst for all four models. Pretraining on natural images therefore does not remove the shift between events, which is consistent with the benchmark's finding that the dominant difficulty is the data rather than the network [4].

E. Cross-dataset evaluation

Table V and Fig. 6 show the post-event-only models in both directions. In domain, the Sen1Floods11 model reaches IoU

TABLE III
FRESH TEST (10 UNSEEN EVENTS, 10,839 PATCHES, MINIMUM AREA 500 PX). SINGLE = MEAN ± SD OVER 3 SEEDS; ENS. = 3-SEED ENSEMBLE.

Model	Single IoU	Ens. IoU	Ens. F1	Ens. precision / recall	Ens. IoU w/o 500 px rule	IoU if no FP on flood-free
U-Net, pre1 + post	0.586 ± 0.019	**0.599**	0.750	0.709 / 0.795	0.593	0.666
`mt_g2` (proposed)	0.587 ± 0.016	0.597	0.747	0.700 / 0.801	0.590	0.672

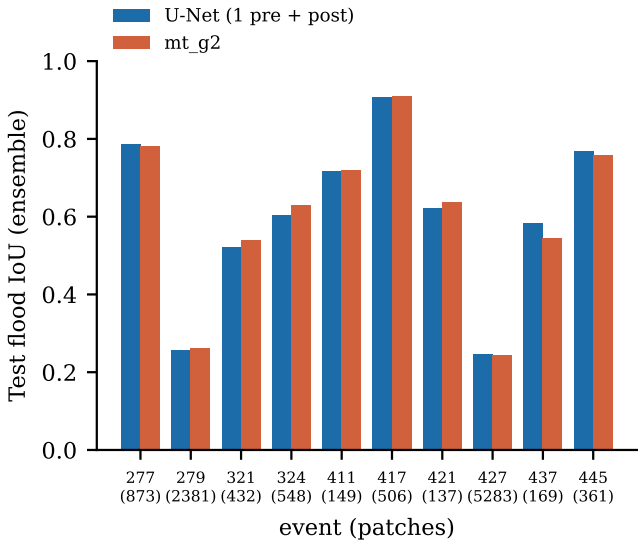


Fig. 4. Per-event test IoU (ensembles). Events 279 and 427 contain 71% of the patches and almost no flood.

TABLE IV
RECENT BASELINES (PRE1 + POST INPUT). VALIDATION: MEAN ± SD OVER 3 SEEDS AND ENSEMBLE. FRESH TEST (POST HOC, EXPLORATORY): SAME. IOU WITH THE 500 PX RULE.

Model	Validation Single	Validation Ens.	Fresh test (post hoc) Single	Fresh test (post hoc) Ens.
U-Net	0.684 ± 0.008	0.691	0.586 ± 0.019	0.599
`mt_g2` (proposed)	0.669 ± 0.004	0.677	0.587 ± 0.016	0.597
U-Net, ResNet-50 enc.	0.693 ± 0.007	**0.703**	0.587 ± 0.021	**0.606**
SegFormer-B2	$\mathbf{0.696 \pm 0.003}$	0.700	0.583 ± 0.005	0.589

0.524 ± 0.012 on the test chips (0.661 for water including permanent water) and 0.608 ± 0.065 on the held-out region Bolivia, so generalization to an unseen region is possible when training and test labels agree. The Kuro Siwo model, applied zero-shot, drops to 0.233 ± 0.124 on the Sen1Floods11 test chips and is unstable on Bolivia (0.230 ± 0.371 over seeds; the ensemble scores 0.073, i.e. one seed works and two fail); uniting its prediction with the JRC layer raises the test water IoU to 0.473 ± 0.067, still below the in-domain 0.661. Region-wise, the Kuro Siwo model is competitive only in Nigeria (0.867 vs. 0.861) and Mekong (0.699 vs. 0.841) and close to zero in Paraguay, Sri Lanka, Pakistan, Ghana and Bolivia (Fig. 6). In the other direction, the Sen1Floods11 model keeps part of the performance on Kuro Siwo validation (0.577 ± 0.062 against 0.656 ± 0.016 in domain) and on test

TABLE V
CROSS-DATASET FLOOD IOU OF THE POST-EVENT-ONLY U-NET (PERMANENT WATER IGNORED; MEAN ± SD OVER 3 SEEDS, ENSEMBLE IN PARENTHESES). BOLD: IN-DOMAIN. †POST HOC.

Test set	Kuro-trained	Sen-trained
Sen1Floods11 test	0.233 ± 0.124 (0.263)	$\mathbf{0.524 \pm 0.012}$ (0.533)
Sen1Floods11 Bolivia (new region)	0.230 ± 0.371 (0.073)	$\mathbf{0.608 \pm 0.065}$ (0.635)
Kuro validation	$\mathbf{0.656 \pm 0.016}$ (0.661)	0.577 ± 0.062 (0.591)
Kuro test, shard 1†	$\mathbf{0.599 \pm 0.004}$ (0.605)	0.370 ± 0.168 (0.388)
Kuro test, shard 2†	$\mathbf{0.246 \pm 0.039}$ (0.251)	0.138 ± 0.071 (0.132)

shard 1 (0.370 ± 0.168 against 0.599 ± 0.004) but is weak on shard 2 (0.138 ± 0.071 against 0.246 ± 0.039). Transfer is thus asymmetric: a model trained on 252 hand-labelled water chips transfers to unseen Kuro Siwo events better than a model trained on thousands of Kuro Siwo flood patches transfers to Sen1Floods11. We did not isolate the cause; the similar backscatter statistics (Fig. 1) suggest that label definition (temporary flood versus water) and scene content matter more than calibration, but this remains a hypothesis. The Kuro Siwo test shards are used post hoc, and with 15 Bolivia chips and three seeds the Bolivia estimates are coarse.

VI. ANALYSIS AND DISCUSSION

Where the errors come from. False-positive pixels on flood-free patches are 40% of all false positives on the test (versus 3% on validation); removing them would raise ensemble IoU from 0.599 to 0.666. This failure mode is invisible on validation, where re-weighting to 69% flood-free patches costs about 0.02 IoU, against about 0.09 on the first test shard. IoU also depends strongly on flood size: on validation it is 0.35 for patches with less than 5% flood and 0.75 for patches above 20%.

Permanent water. Permanent water is scored as non-flood. On the `base` ensemble it accounts for 32% (validation), 16% (test shard 1) and 26% (test shard 2) of false-positive pixels, and ignoring it entirely would add only 0.020–0.034 IoU. A three-class model might help, but the upper bound is much smaller than the flood-free-patch effect.

Acquisition gap. The hypothesis that long gaps hurt is not supported: within the first test shard IoU rises with gap, and on the fresh test the proposed model is slightly better in three of four gap bins and on six of ten events, but worse in the 180–240-day bin that holds most patches (Fig. 5). No superiority can be claimed.

Explainability. We analysed the plain U-Net (pre1 + post) and the proposed model on random samples of 1,200 vali-

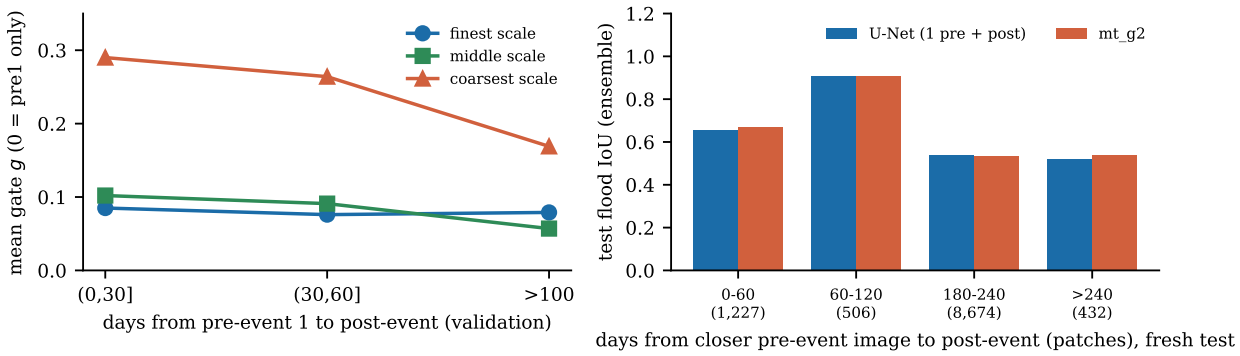


Fig. 5. Left: learned mean gate per acquisition-gap bin (validation, seed 0). Right: test IoU by days from the closer pre-event image to the event (ensembles); patch counts in parentheses.

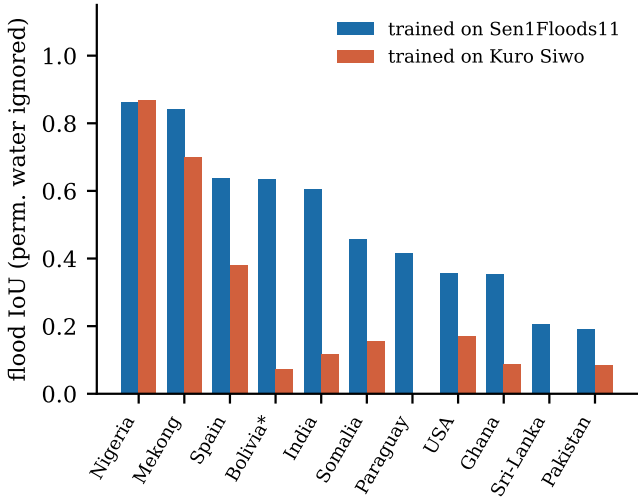


Fig. 6. Per-region IoU on Sen1Floods11 test chips (* = held-out region Bolivia), models trained on each dataset (single post-event image, permanent water ignored).

dation and test patches (first test shard) with ensembles of three seeds. *Input attribution.* We computed exact Shapley values [36] over the four input channels (16 coalitions; an absent channel is set to the training mean) and permutation importance. The post-event VV channel dominates: its Shapley value is 0.469 IoU on validation and 0.455 on test, out of a total of 0.685 and 0.591, against 0.036–0.089 for post VH and 0.044–0.073 for the two pre-event channels; permuting post VV costs 0.54 (validation) and 0.51 (test) IoU, post VH 0.23 and 0.20, and the pre-event channels 0.07–0.11 (Fig. 9). Gradient×input on four individual patches gives the same ordering (post channels 73–82% of the attribution). For the proposed model, with the three dates as players, the post-event image contributes 0.572 (validation) and 0.469 (test), pre-event 1 contributes 0.056 and 0.069, and pre-event 2 contributes 0.046 and 0.057; the older image is therefore not ignored when it is removed, even though the gate weights it little, which shows that its features still enter the model through the shared encoder and the fusion layer. Replacing a channel by its mean is an out-of-distribution input, so these values measure dependence rather than causal importance. *Grad-CAM and gate maps.* For the plain U-Net, Grad-CAM taken from the last decoder block (target: flood evidence over ground-truth flood pixels) covers the flooded areas in the good events 417 and 277 and part of the flooded area in event 279, where it also highlights textured vegetation that the model wrongly labels flood (Fig. 8); in event 427, which scores IoU 0.02, there is almost no activation, and the post-event VV image shows no visibly darker flooded area, i.e. the model finds no evidence of flood. The bottleneck Grad-CAM is coarse and less specific. The spatial gate of the proposed model is largest along field boundaries and edges at the finest scale and takes very different values at the coarsest scale (inside versus outside flood: 0.13 vs. 0.00 in event 417, 0.46 vs. 0.81 in event 277, 0.58 vs. 0.26 in event 279, 0.05 vs. 0.06 in event 427), so there is no consistent link between the gate and flooded pixels (Fig. 10). *Qualitative errors.* Fig. 7 shows the ensembles on typical patches: in event 279 false alarms (red) occur next to true flood on rough surfaces, and in event 427 the whole flood is missed (blue) by both models, the largest source of error in the test.

Limitations. (i) Pre-event models were evaluated on one dataset, Kuro Siwo; Sen1Floods11 has no pre-event pair, so only post-event-only models were compared across datasets, with differing label definitions (flood versus water) handled by ignoring permanent water, which is an approximation. The Bolivia estimates rest on 15 chips. (ii) Only three validation events, so differences below about 0.02 IoU on validation are not reliable. (iii) Event locations cannot be derived from the metadata, so only generalization to unseen events is supported; events 277 and 427 occur in two test shards. (iv) The pretrained baselines were trained with one recipe and learning rate and no tuning, and they were evaluated on the test post hoc; a SAR-pretrained foundation model (Prithvi, TerraMind) was not tested. (v) Label noise was not examined, and the failure of events 279 and 427 was not traced to a cause. (vi) The training data are dominated by a few large events, and a single recipe

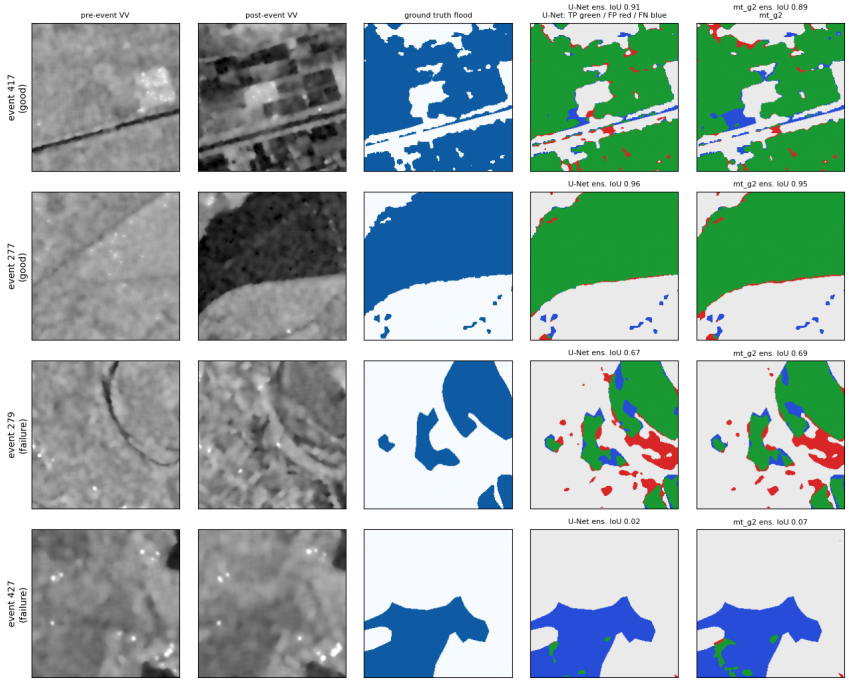


Fig. 7. Predictions of the two ensembles on one fresh-test patch per event: events 417 and 277 (good), 279 and 427 (failures). Green = true positive, red = false positive, blue = false negative, grey = invalid. Patch-level IoU in the titles (without the 500 px rule).

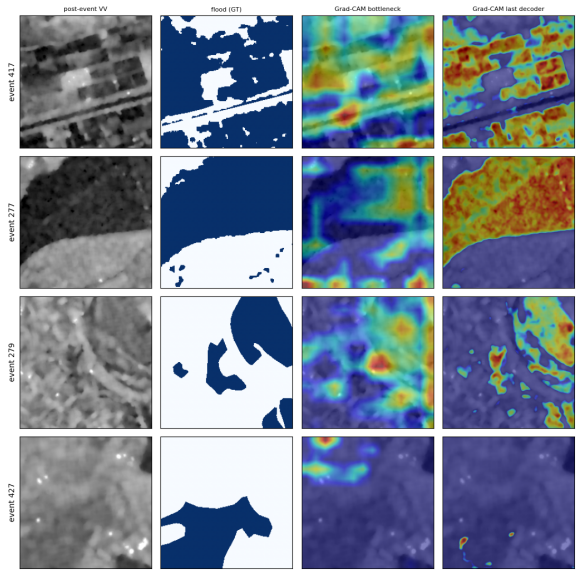


Fig. 8. Grad-CAM of the plain U-Net (seed 0) on the same patches, bottleneck and last decoder block.

was used for all models. (vii) Shapley values replace a channel by the training mean, and Grad-CAM was inspected on four patches, so the explanations are illustrative.

Future work. Training with long-gap augmentation (use the older image as "pre" as well), a patch-level flood/no-flood head to cut false alarms, three-class training, joint training on both datasets with a water/flood label mapping, SAR-pretrained foundation models, and more validation events or

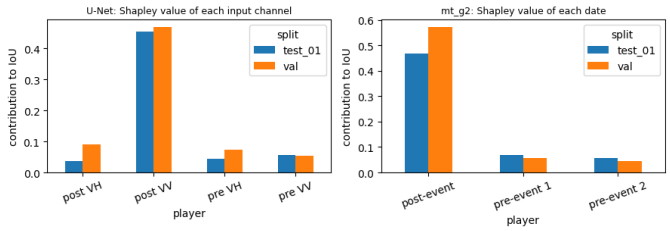


Fig. 9. Shapley value (contribution to IoU) of each input channel of the plain U-Net (left) and of each date for the proposed model (right), on validation and first test shard.

event-grouped cross-validation.

VII. Conclusion

On Kuro Siwo with strictly event-disjoint evaluation, one pre-event Sentinel-1 image is what matters: it adds 0.15 IoU over post-event data alone. A second pre-event image, temporal attention, gated and gap-aware fusion, terrain channels and speckle filtering gave no measurable gain over a plain U-Net, and the proposed multi-temporal model matched but did not beat it on ten unseen test events; pretrained ResNet-50 and SegFormer-B2 baselines gained about 0.01 IoU on validation and nothing measurable on the test events. Post-event-only models transfer poorly between Kuro Siwo and Sen1Floods11, and attribution analyses show that the post-event VV channel carries most of the prediction. The remaining error is dominated by false alarms on flood-free and small-flood patches and by event-to-event shift, so progress is more likely to come from data and training design, such as flood-free sampling, label

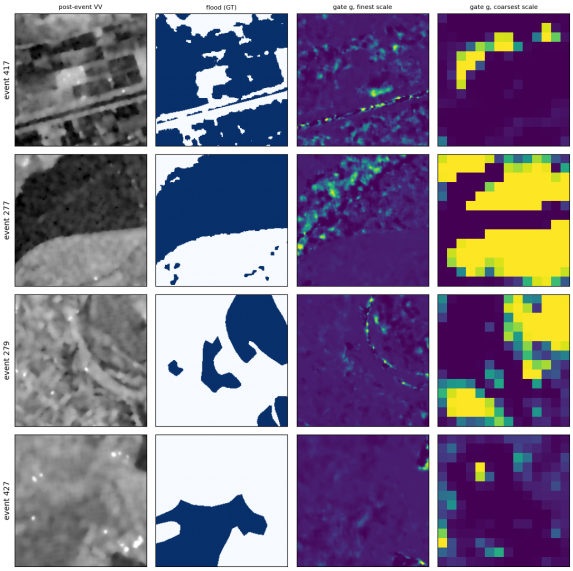


Fig. 10. Spatial gate g of the proposed model at the finest and coarsest scale (0 = only pre-event 1).

harmonisation across datasets and long-gap augmentation, than from architecture.

Reproducibility

Code is in a single Kaggle notebook https://github.com/Vasanth-Loganathan/flood-mapping; seeds, splits (event lists), decision rules and the frozen test configuration are recorded in the project files; Sen1Floods11 is read from its public Google Cloud bucket.

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